

1 Deriving Continuity Equation

Beginning with a control volume, as shown below, it is possible to derive a continuity equation for mass balance.

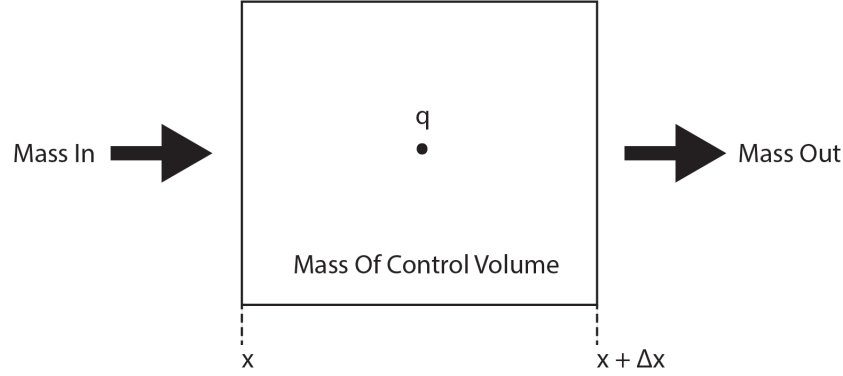


Figure 1: Control volume defined to develop mass balance equation.

Because mass can neither be created or destroyed, the mass conservation for this control volume can be described as

$$\underbrace{\text{Mass Final} - \text{Mass Initial}}_{\text{Accumulation}} = \underbrace{\text{Mass In} - \text{Mass Out}}_{\text{Convection}} + \underbrace{\text{Generation}}_{\text{Source}}, \quad (1)$$

which can be stated as

$$\underbrace{m \cdot V \Big|_{t+\Delta t} - m \cdot V \Big|_t}_{\text{Accumulation}} = \underbrace{F \cdot \Delta t \cdot A \Big|_x - F \cdot \Delta t \cdot A \Big|_{x+\Delta x}}_{\text{Convection}} + \underbrace{q \cdot \Delta t \cdot V}_{\text{Source}}. \quad (2)$$

Here, m is the mass density of the control volume, F is the mass flux, V is the volume of the control volume, Δt is the change in time, and A is the cross-sectional area orthogonal to the direction of Δx . By dividing this equation by the Δt and V , one obtains

$$\frac{m}{\Delta t} \Big|_{t+\Delta t} - \frac{m}{\Delta t} \Big|_t = \frac{F}{\Delta x} \Big|_x - \frac{F}{\Delta x} \Big|_{x+\Delta x} + q, \quad (3)$$

which can be reformulated as

$$\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q. \quad (4)$$

Next, the limit is taken as Δt and Δx go to zero, i.e.,

$$\lim_{\substack{\Delta t \rightarrow 0 \\ \Delta x \rightarrow 0}} \left(\frac{m \Big|_{t+\Delta t} - m \Big|_t}{\Delta t} = \frac{F \Big|_x - F \Big|_{x+\Delta x}}{\Delta x} + q \right), \quad (5)$$

resulting in

$$\frac{\partial m}{\partial t} = -\frac{\partial F}{\partial x} + q. \quad (6)$$

Note that this equation, representing the conservation of total mass, is also called the Continuity Equation. In general form, the equation can be stated as

$$\frac{\partial m}{\partial t} = -\nabla \cdot F + q. \quad (7)$$

The m (mass density of fluid per bulk volume) of the left term can be replaced like

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot F + q. \quad (8)$$

The F (mass flux in units of $\frac{kg}{m^3} \frac{m}{sec}$) can be replaced like

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot (\rho \cdot u) + q. \quad (9)$$

In this formulation, there are no source terms (wells), such that,

$$\frac{\partial (\phi \rho)}{\partial t} = -\nabla \cdot (\rho \cdot u). \quad (10)$$

Note further that the fluid is incompressible and porosity is constant in time, such that the accumulation term vanishes,

$$0 = -\nabla \cdot (\rho \cdot u). \quad (11)$$

Next, by assuming that fluid transport is entirely governed by pressure-driven advection (diffusion is neglected), and that fluid velocity can be described by Darcy's Law, $u = -\frac{k}{\mu} \nabla P$, the continuity equation is given by

$$-\nabla \cdot \left(\rho \left(-\frac{k}{\mu} \nabla P \right) \right) = 0, \quad (12)$$

which simplifies to,

$$\nabla \cdot \left(\rho \frac{k}{\mu} \nabla P \right) = 0, \quad (13)$$

Considering that density and viscosity are constant,

$$\frac{\rho}{\mu} \nabla \cdot (k \nabla P) = 0. \quad (14)$$

Therefore, the continuity equation is,

$$\nabla \cdot (k \nabla P) = 0. \quad (15)$$

2 Finite Volume Method Discretization

Beginning with the continuity equation from mass balance, Equation 15,

$$\nabla \cdot \left(\frac{k}{\mu} \nabla P \right) = 0, \quad (16)$$

with μ included for convenience, the equation is integrated over a control volume V_i ,

$$\int_{V_i} \nabla \cdot \left(\frac{k}{\mu} \nabla P \right) dV = 0. \quad (17)$$

Applying the divergence theorem converts the volume integral into a surface integral,

$$\int_{A_i} \frac{k}{\mu} \nabla P \cdot \mathbf{n} dA = 0, \quad (18)$$

where $\mathbf{n}$ is the outward-pointing unit normal vector.

For a one-dimensional domain, the fluxes through the left and right faces of the control volume can be approximated using finite differences,

$$A \left(\frac{k_{i+\frac{1}{2}}}{\mu} \frac{P_{i+1} - P_i}{\Delta x} \right) - A \left(\frac{k_{i-\frac{1}{2}}}{\mu} \frac{P_i - P_{i-1}}{\Delta x} \right) = 0, \quad (19)$$

where A is the cross-sectional area normal to the flow direction and i indicates cell number.

Introducing the transmissibility,

$$T_{i+\frac{1}{2}} = \frac{k_{i+\frac{1}{2}} A}{\mu \Delta x}, \quad (20)$$

the discretized equation becomes

$$-T_{i-\frac{1}{2}} P_{i-1} + \left(T_{i-\frac{1}{2}} + T_{i+\frac{1}{2}} \right) P_i - T_{i+\frac{1}{2}} P_{i+1} = 0. \quad (21)$$

Note that the edge permeability, $k_{i+\frac{1}{2}}$, is the harmonic average of the two adjacent cells' permeabilities, $k_{i+\frac{1}{2}} = \frac{2k_i k_{i+1}}{k_i + k_{i+1}}$, guaranteeing continuity of flux across material interfaces. Note further that, at the constant-pressure (Dirichlet) boundaries, the boundary transmissibility is

$$T_B = \frac{k_i A}{\mu \frac{\Delta x}{2}}, \quad (22)$$

and Equation 21 reads,

$$-T_B P_L + \left(T_{1+\frac{1}{2}} + T_B \right) P_1 - T_{1+\frac{1}{2}} P_2 = 0, \quad -T_{N_x-\frac{1}{2}} P_{N_x-1} + \left(T_{N_x-\frac{1}{2}} + T_B \right) P_{N_x} - T_B P_R = 0, \quad (23)$$

for the left and right boundaries, respectively. Here, N_x is the number of cells in the domain and P_L and P_R are the left and right pressure boundary conditions, respectively. These equations can be rewritten as,

$$\left(T_{1+\frac{1}{2}} + T_B \right) P_1 - T_{1+\frac{1}{2}} P_2 = T_B P_L, \quad -T_{N_x-\frac{1}{2}} P_{N_x-1} + \left(T_{N_x-\frac{1}{2}} + T_B \right) P_{N_x} = T_B P_R, \quad (24)$$

to facilitate their incorporation in the linear system of equations. Indeed, the discretized equations may be assembled into a linear system,

$$\mathbf{A}\mathbf{P} = \mathbf{q}, \quad (25)$$

where $\mathbf{A}$ is the transmissibility matrix based on Equation 21, $\mathbf{q}$ contains the contributions from the prescribed pressure boundary conditions, and where $\mathbf{P}$ is the vector of unknown cell-center pressures,

$$\begin{bmatrix} T_B + T_{1+\frac{1}{2}} & -T_{1+\frac{1}{2}} & 0 & \cdots & 0 \\ -T_{2-\frac{1}{2}} & T_{2-\frac{1}{2}} + T_{2+\frac{1}{2}} & -T_{2+\frac{1}{2}} & \cdots & 0 \\ 0 & -T_{3-\frac{1}{2}} & T_{3-\frac{1}{2}} + T_{3+\frac{1}{2}} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & -T_{N_x-\frac{1}{2}} & T_{N_x-\frac{1}{2}} + T_B \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ \vdots \\ P_{N_x} \end{bmatrix} = \begin{bmatrix} T_B P_L \\ 0 \\ 0 \\ \vdots \\ T_B P_R \end{bmatrix}. \quad (26)$$

The system of equations is then solved as,

$$\mathbf{P} = \mathbf{A}^{-1}\mathbf{q}. \quad (27)$$

Note that the interface transmissivities $T_{2-\frac{1}{2}} = T_{1+\frac{1}{2}}$, for example. They are written out this way for clarity. The fluid velocity across the interfaces can then be found using Darcy's Law as,

$$u = -\frac{k}{\mu} \frac{\partial P}{\partial x}. \quad (28)$$

Applying a finite-difference approximation at interface $i + \frac{1}{2}$ gives

$$u_{i+\frac{1}{2}} = -\frac{k_{i+\frac{1}{2}}}{\mu} \left(\frac{P_{i+1} - P_i}{\Delta x} \right). \quad (29)$$

Note that for the domain boundaries these velocities are,

$$u_L = -\frac{k_1}{\mu} \left(\frac{P_1 - P_L}{\frac{\Delta x}{2}} \right), \quad u_R = -\frac{k_{N_x}}{\mu} \left(\frac{P_R - P_{N_x}}{\frac{\Delta x}{2}} \right). \quad (30)$$